EPFL - Computer Science

COM-407 Advanced Networks

Lecture 04 — October 7, 2025

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IPv4 IPv6 Final

MAC Address Resolution (IP address → MAC address)

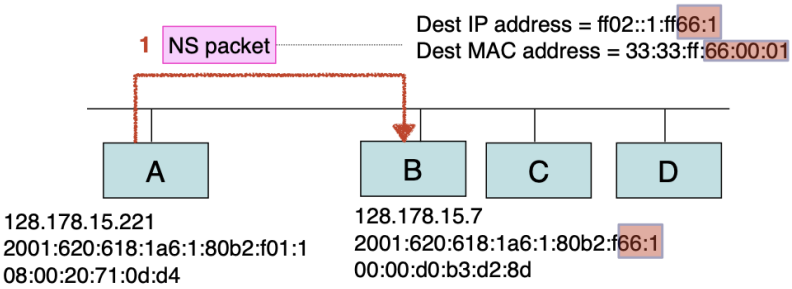
Why?

- Say A has a packet to send to a next hop B inside the same subnet, either final destination or default gateway router;
- A knows only B's IP address, and must find B's MAC address, as within a subnet we use the MAC layer to move data

How?

- A broadcasts (in IPv4) or multicasts (in IPv6) a packet to the LAN asking: "who has the specific IP address of B?"
- All hosts listening to that IP address (in principle only B) respond back with their MAC address

MAC address resolution in IPv6: Neighbor Discovery Protocol—NDP



Steps:

i. A sends a Neighbor Solicitation (NS) packet with the question: "who has IP address B?"

- NS's dest. IP address = a special multicast address (Solicited Node Multicast Address) ⇒ last 24 bits are copied by B's IP address

- NS's dest. MAC address is derived from the multicast address in a similar way

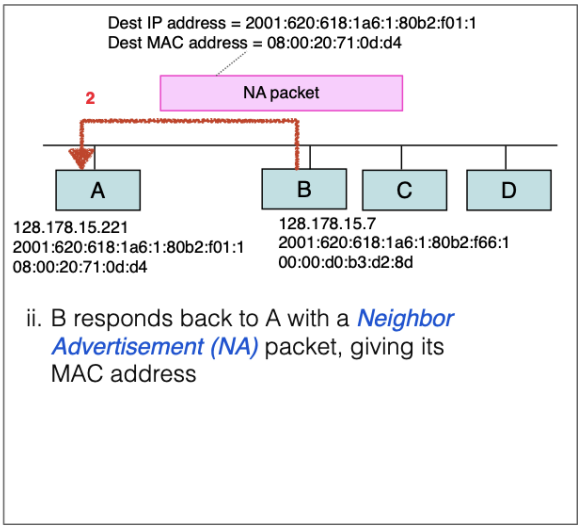
ii. B responds back to A with a Neighbor Advertisement (NA) packet, giving its MAC address

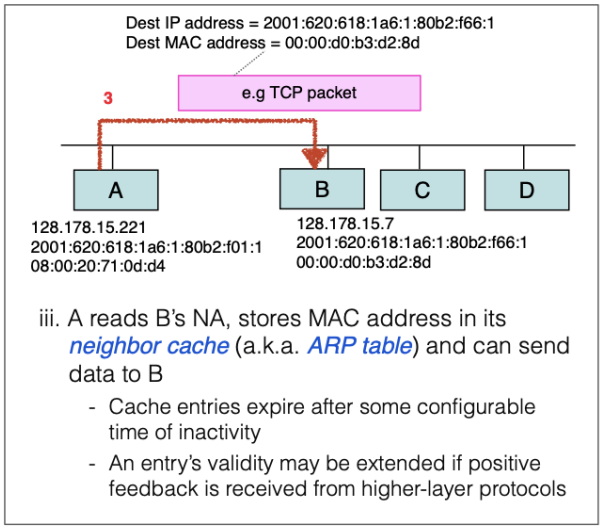
iii. A reads B's NA, stores MAC address in its neighbor cache (a.k.a ARP table) and can send data to B

- Cache entries expire after some configurable time of inactivity
- An entry's validity may be extended if positive feedback is received from higher-layer protocols

NA, NS packets are carried as ICMPv6 packets

encapsulated inside IPv6 packets with next-header field = 58 (0x3a)



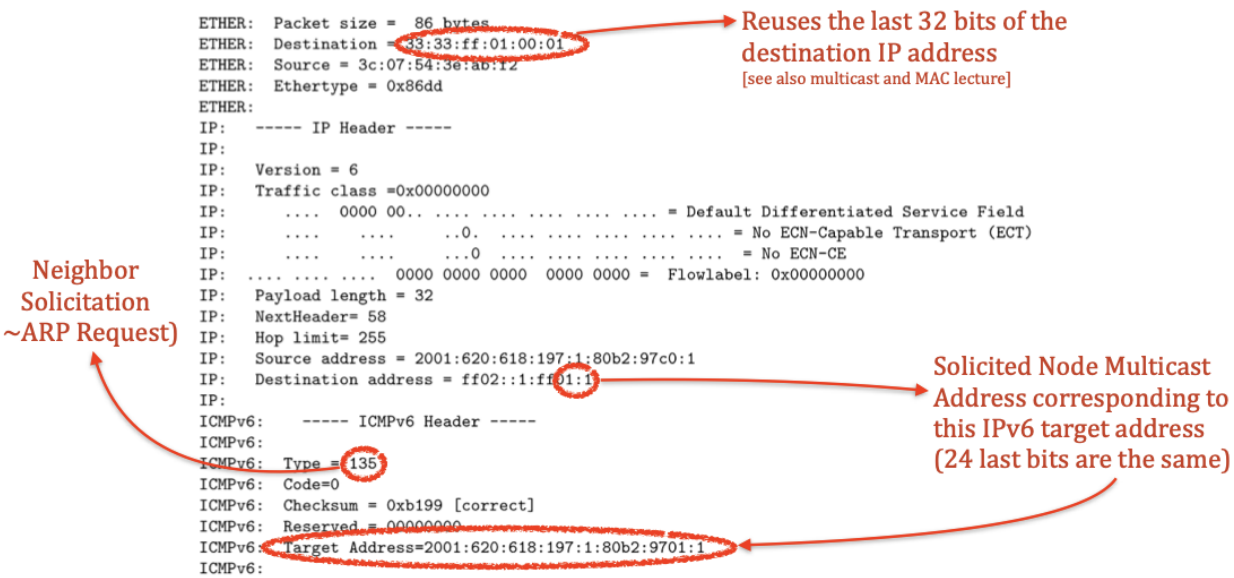


The Solicited Node Multicast Address in detail

- Obtained by adding last 24 bits of target IP address to the following prefix: ff02::1:ff00:0/104
- A packet with such a destination address is received by all nodes all hosts whose IP address has the same last 24 bits (all of them listen/"subscribe" to this multicast address)
- Used only in IPv6 NDP or other protocols; IPv4 uses broadcast instead

Target address	Compressed	2001:620:618:1a6:1:80b2:f66:1
	Uncompressed	2001:0620:0618:01a6:0001:80b2:0f66:0001
Solicited Node multicast address	Uncompressed	ff02:0000:0000:0000:0000:0001:ff66:0001
	Compressed	ff02::1:ff66:1

Look into an NDP Neighbor Solicitation Packet



MAC Address Resolution with IPv4

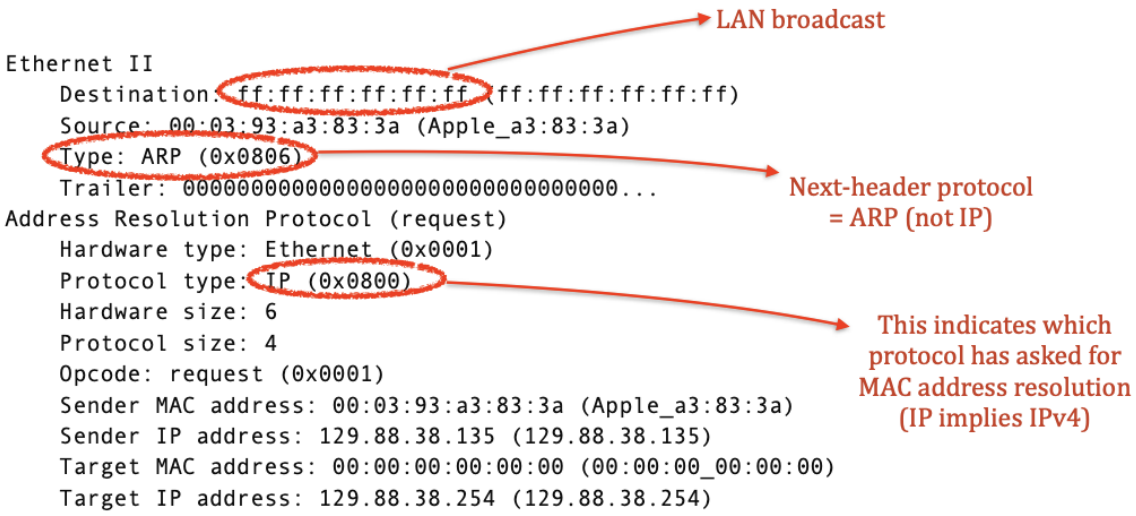
Similar to NDP, except:

- Terminology:
  - the protocol is called ARP (Address Resolution Protocol)
  - NS/NA pkts of NDP are called ARP Request/ ARP reply
- Protocol type
  - ARP pkts are not ICMP packets (not even encapsulated in IP packets)
- Reachability:
  - ARP request is broadcast to all nodes in LAN (instead of multicast)

Question: Why don't we do the same like DNS? Having a server that host all the data for MAC resolution address?

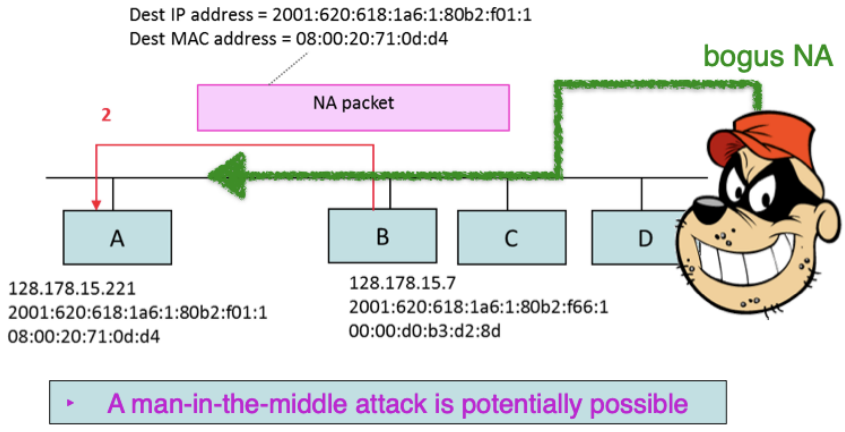
Solution: The main reason is that IP address may be reused. If we had a server hosting this information, the server would have to update its entries every time there's changes.

Look into an ARP request



Security Issues with ARP / NDP

- ARP spoofing: ARP replies and NAs can be forged
  - attacker intercepts packets destined to B ——¿ easier in IPv4, bc ARP requests are broadcast
  - sends a bogus NA/ARP reply with its own MAC address
  - may cause A to map their MAC address to B's IP address, if the bogus NA arrives faster at A



How to prevent ARP spoofing in LANs?

- **DHCP snooping**: switch/WiFi base station observer all DHCP traffic and remembers mappings IP addr ⇔ MAC addresses
- **Dynamic ARP inspection**: switch filters all ARP (or NDP) traffic and allows only valid answers – removes invalid broadcasts (IPv4) and multicasts (IPv6)
- Such solutions are deploy in enterprise networks, rarely at home or WiFI access points

The MAC Layer

Contents

1. MAC as Shared Medium
2. Switches
3. Frame format and MAC addresses
4. Virtual LANs
5. WiFi Distribution Systems
6. Security Aspects

1. MAC: Medium Access Control

The MAC layer is responsible for sharing data between the LAN

Why invented? So that multiple LAN devices share a single physical medium:

- cable (old/coxial Ethernet, Powerline); or
- wireless radio link (GSM, WiFI, Bluetooh)

What is the problem?

If several systems transmit at the same time, several frames will appear on the transmission medium (channel) and they will collide/interfere

Solution 1: joint decoding (allow everyone to talk and decode) – used in cellular networks (e.g. CDMA)

Solution 2: mutual exclusion (only one system can talk at a time), ideally via a distributed protocol

How to implement mutual exclusion?

Deterministic protocols

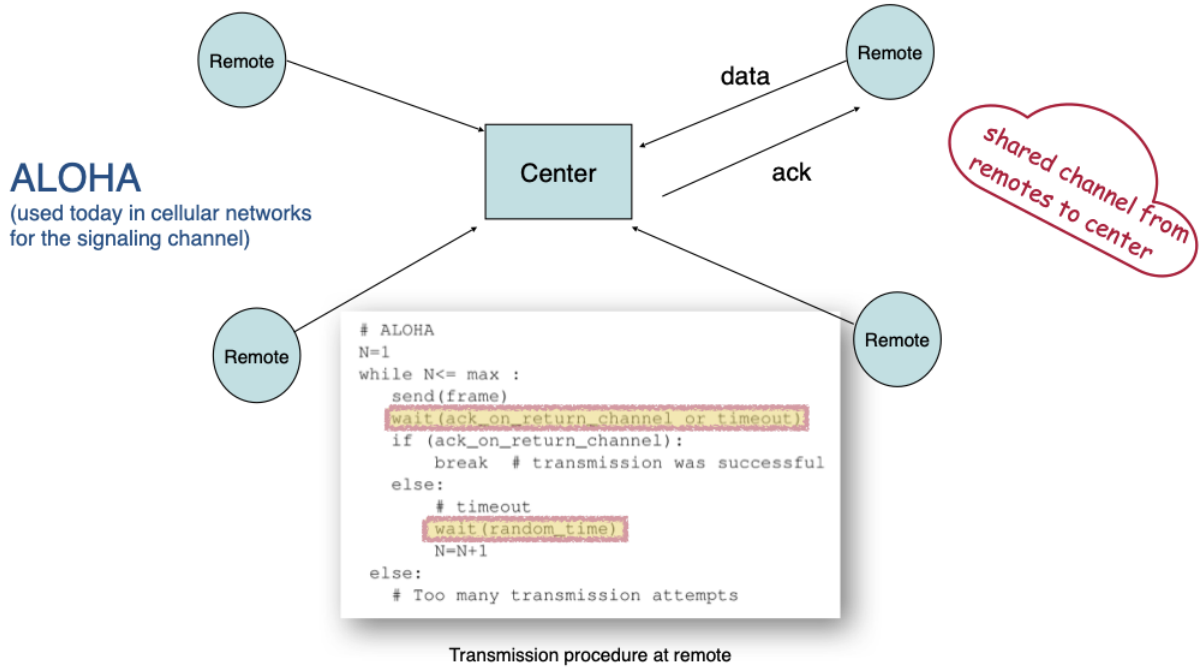
- use an arbitration for accessing the transmission medium(channel)
  - Time Division Multiple Access (**TDMA**); each host is given a **dedicated timeslot** to transmit
  - Frequency Division Multiple Access (**FDMA**): each host is given a **dedicated frequency band** to use
  - CAN bus: the **highest-priority** transmitting host dominates other transmission
  - Token ring, FDDI: every host takes turn and passes a token to next host; whoever has the token, can transmit
    - more flexible than TDMA and FDMA, may achieve higher throughput
- avoid collisions by construction, proactively
- but... work only with few users, as they require synchronization and pre-configuration (if a host goes down or a new one appears, we need to take the network down and reconfigure it)

Stochastic (random access) protocols

- allow everyone to transmit at any time, detect collisions, and react
- host transmits at full rate and if a collisions is detected, they retransmit after some random time ("back-off")
  - WiFi, coaxial/historical Ethernet, Zigbee, signaling channel on cellular networks, Ethernet over Powerline

Principles of random-access protocols

ALOHA

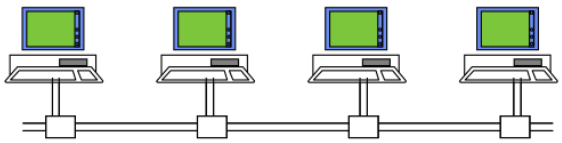


- Two channels: uplink and downlink. Shared channel from remotes to center. The downlink is easy, the center (radio-station) can schedule it's transmission.
- The others remotes need to schedule their transmission. If they all transmit in the same time, they will collide.
- The maximum utilization is 18% of the channel capacity, which is very small. This is assuming independent and exponential inter-arrival times of pkts and retransmissions.

CSMA (carrier sense multiple access)

- Before you transmit, simply listen to the channel. If it's busy, just wait until it's free. It assumes a single transitive channel (everyone can hear everyone)
- drastically improves efficiency over Aloha
- In CSMA, many collisions can be avoided, but not all. Two or more stations may sense that the medium is idle and start transmitting at time instants that are close enough for a collision to occur (propagation delay is also in the source of this problem)

```
# persistent CSMA
N=1
while N<= max :
  listen(channel)
  wait(channel_becomes_free)
  send(frame)
  wait(ack or timeout)
  if ack :
       break  # transmission was successful
  else :
       # timeout
       N=N+1
# end of while loop
  # Too many transmission attempts
```



CSMA works better if: propagation delay is small

Non-persistent CSMA

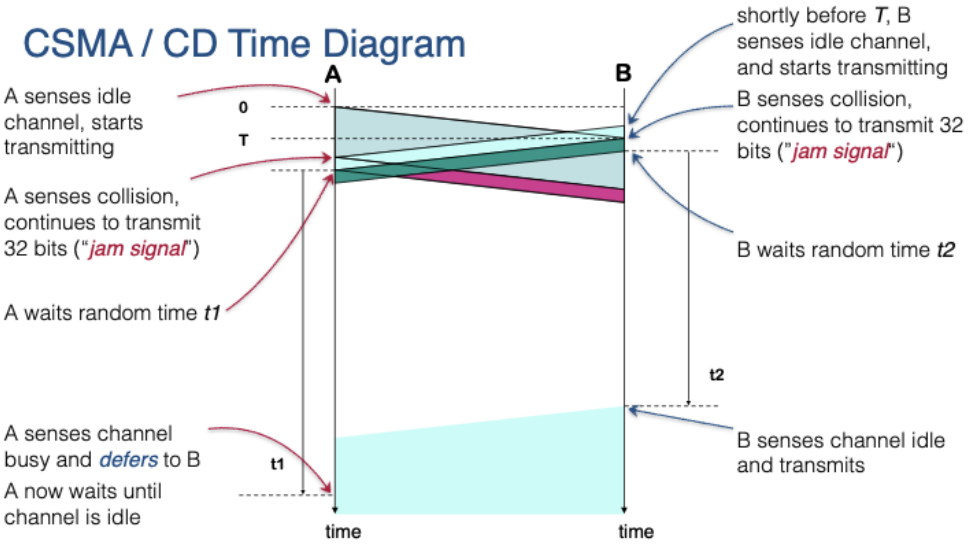
- uses random waiting times if channel is busy
- avoid hosts' synchronization
  - if all stations choose the random delays independently, and
  - if the waiting times are larger than the propagation delay
  - then, with high probability, only one of the transmitting stations seizes the channel
- improves utilization

CSMA / CD (Collision Detection) detects collisions as they occur

ACKs replaced by CD: channel is monitored even during transmission to detect collisions

```
# CSMA/CD pseudo-code
N=1
while N<= max :
   wait(channel_becomes_free)
   send(frame)
   wait_until (end_of_frame) or (collision)
   if collision detected:
        stop transmitting
        send(jamming)
        k = min (10, N)
        r = random(0, 2^k - 1) * slotTime
        wait(r*slotTime)
        N=N+1
   else :
        wait(inter-frame_delay)
        break
 # end of while loop
   # Too many transmission attempts
```

Exponential back-off



A Minimum Frame Size is needed to guarantee collision detection

To avoid undetected collisions, we impose a minimum frame size that depends on the link's bit rate and round-trip propagation delay (a.k.a slotTime)

transmission delay = frameSize/bitRate > slotTime ⇒ frameSize > slotTime*bitRate

bandwidth-delay product

CSMA / CD Summary

- CSMA/CD improves on CSMA by requiring that stations detect collisions and stop transmitting (after 32 bits, called **jam** bits, in order to ensure that all systems properly recognize the presence of collisions).
- The jam bits (or jam signal) is a 32-bit binary pattern indicating that a collision has occurred.
- CSMA/CD has a *better performance* than Aloha or CSMA
- After a collision is detected, stations will re-attempt to transmit after a random time
- The random time before retransmission is chosen in such a way that if repeated collisions occur, then this time increases exponentially (exponential backoff). The downside of exponential backoff is that: in case of congestion (too many collisions), the access to the channel is slowed down
- Acknowledgements are not necessary because absence of collision means that the frame can be transmitted
- An inter-frame delay ("gap") of 9.6μs is used to avoid blind times, during which adapters are filtering typical noise at transmission ends.

CSMA/CD = early Ethernet (coax cables and hubs)

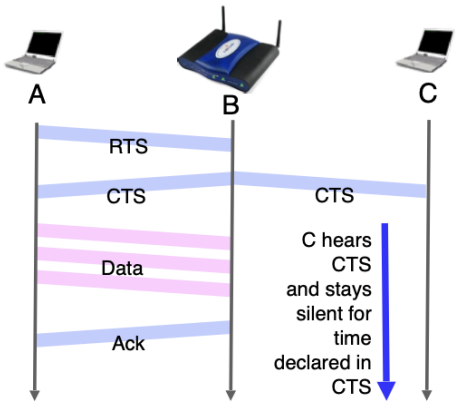
Early Ethernet (half-duplex) addresses collisions by using:

- CSMA/CD
- with restrictions about the slotTime (hence also the min frame size)
  - e.g., 10Mbps half-duplex Ethernet uses slotTime = 51.2ms $\Rightarrow$ min frame size = 64Bytes

Wi-Fi = CSMA/CA (Collision Avoidance) = variant of CSMA/CD

Uses:

- ACKs to detect collisions (i.e. if they are absent)
- Carrier sensing, but does not always work:
  - e.g., C may be "hidden"from A;
  - and this is *mitigated* through RTS/CTS
- Further optimizations to avoid collisions:
  - more conservative channel backoff. Even if the channel is `idle`, still wait for random time before transmitting
  - network allocation vector. The MAC header contains the transmission time needed for the frame; it its a form of virtual carrier sensing



Question In a WiFi network with a single channel and a single base station, how many data frames can be transmitted concurrently?

A. At most one in total
B. One from mobile to base station and one from base station to mobile
C. At most one per mobile to base station and one from base station
D. At most one per mobile to base station and one per mobile from base station
E. I don't know

Solution: A. The WiFi MAC layer controls the access of all senders to the radio channel. So, it makes sure that only one can transmit at a time.